ANDRITZ Hydro

Installation and Commissioning

Inspection and Test Plan

Wet-Commissioning

Document #: ITP-GMS- 1520
Project: GM Shrum

Revision : 3
Unit #:

ITP step #	Pre-requisite	Description	Responsibilities: Perform	Responsibilities: Witness	Safety risk?	Procedure #	Procedure step #	Form	Date and signatures: Date Performed	Date and signatures: Witness 1	Date and signatures: Witness 2	Comments
Pre-Com												
1520 - 1	N/A	Wiring checks	AH	BCH	YES	P-1500-01	Section # 1	Q-1500-000				
1520 - 2	N/A	RTD Checks	AH	BCH	YES	N/A	N/A	Q-1500-100				
1520 - 3	N/A	Air Gap sensors verification	AH	BCH	YES	N/A	N/A	Q-1500-430F				
UNIT WALKDOWN												
1520 - 4	N/A	Turbine walk down	BCH	AH	YES	N/A	N/A	N/A				
1520 - 5	N/A	Generator walk down	AH / BCH	BCH	YES	N/A	N/A	N/A		WC		
Mechanical static tests												
1520 - 6	1 and 2	Unit water-up - Leakage verifications	BCH	AH	YES	BCH Procedure	N/A	N/A				
1520 - 7	3	All cooling water system verification	BCH	AH	YES	BCH Procedure	N/A	N/A				
Wet Tests - No-load, unexcited												
1520 - 8	1 to 7	UNIT CLEARED FOR FIRST ROTATION	AH	BCH	YES	N/A	N/A	N/A	JAN 21st 2013	WC		
1520 - 9	8	Initial Rotation	AH / BCH	BCH	YES	P-1522-01	7.4	Q-1522-010	JAN 21st 2013	WC		
1520 - 10	9	Balancing and bearing heat run up to nominal speed	AH	BCH	YES	P-1522-01	7.5	Q-1522-010 Q-1522-140	JAN 22-24TH 2013	WC		

ANDRITZ Hydro

Document #: ITP-GMS- 1520
Project: GM Shrum

Installation and Commissioning
Inspection and Test Plan
Wet-Commissioning

Revision : 3
Unit #:

ITP step #	Pre-requisite	Description	Responsibilities: Perform	Responsibilities: Witness	Safety risk?	Procedure #	Procedure step #	Form	Date and signatures: Date Performed	Witness 1	Witness 2	Comments
1520 - 11	10	Mechanical overspeed test	AH / BCH	BCH	YES	P-1522-01 P-1522-14	7.6 to 7.8 Section # 6	Q-1522-030A Q-1522-140	JAN 24TH 2013	WC	2013-03-20	MECH OR IDX OS TO BE REPRINTED DONE FEB 19/13
1520 - 12	11	Inspection after mechanical runs (Rotating parts inspection)	AH / BCH	BCH	YES	P-1522-01	7.9	Q-0000-001	JAN 31ST 2013	WC	2013-03-20	
1520 - 13	11	Rotor polarization index test	AH	BCH	YES	P-1510-03	Section # 9	Q-1510-030	FEB 1ST 2013	WC	2013-03-20	
Wet Tests - No-load, excited												
1520 - 14	11	Connect Shorting bar & temporary excitation supply	BCH	AH	YES	BCH Procedure	N/A	N/A	FEB 1-4TH 2013	WC		Shorting bar supply by BCH
1520 - 15	12 to 14	Short-Circuit saturation test	AH / BCH	BCH	YES	P-1523-01	Section # 6	Q-1523-010	FEB 5TH 2013	WC	2013-03-20	
1520 - 16	15	Remove shorting bar	BCH	AH	YES	BCH Procedure	N/A	N/A	FEB 6TH 2013	WC		
1520 - 17	16	Open circuit saturation test	AH / BCH	BCH	YES	P-1523-02	Section # 6	Q-1523-020	FEB 7TH 2013	WC	2013-03-20	
1520 - 18	16	Waveform deviation factor measurement	AH	BCH	YES	P-1543-01	Section # 6	Q-1543-010	FEB 7TH 2013	WC	2013-03-20	Only first unit
1520 - 19	17 and 18	Connect permanent excitation supply	BCH	AH	YES	BCH Procedure	N/A	N/A	FEB 8-10TH 2013	WC		
Load runs												
1520 - 20	19	Heat run	AH / BCH	BCH	YES	P-1524-04	Section # 6	Q-1524-040	FEB 17-19TH 2013	WC	2013-03-20	Rotor temperature only.

ITP step #	Pre-requisite	Description	Responsibilities		Safety risk?	Procedure #	Procedure step #	Form	Date and signatures			Comments
			Perform	Witness					Date Performed	Witness 1	Witness 2	
Completion												
1520 - 21	20	Remove commissioning equipment	AH	N/A	YES	N/A	N/A	N/A	FEB 20th 2013	WC		

Legend

Company	*Short*	*Responsibilities*	*Comments*					
Andritz Hydro	AH	Rotor poles						
British Columbia Hydro	BCH	Customer						

Revision summary

Revision 3
Prepared by: Marc-André Pilon, T.P. Date: 10-Jan-2013
Approved by: François Sergerie
- General revision

Revision 2
Prepared by: Marc-André Pilon, T.P. Date: 10-Aug-2012
Approved by: François Sergerie
- Added QCR-1522-140 at step 1520-10.

Revision 1
Prepared by: Marc-André Pilon, T.P. Date: 2-Jul-2012
Approved by: François Sergerie
- General revision following customer comments.

Revision 0
Prepared by: Marc-André Pilon, T.P. Date: 2-May-2012
Approved by: François Sergerie

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: ________

Revision # 00

Project: BCH GMS

Unit : 4

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Andritz Hydro will not submit the QCR's that correspond to the following ITP steps as they were deemed not applicable (N/A): 1520-4, 1520-6, 1520-7, 1520-8, 1520-14, 1520-16, 1520-19, 1520-21

NCR #: N/A

Measurement Instruments	T°	Measured by:	14-Mar-13
N/A	Before:		
	After:	ANDRITZ Rep. Kirittopoulos K.	14 Mar. 13
		Cust. Repres.:	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 1520-1

Revision # 00

Project: BCH GMS

Unit : 4

Page 1 of 1

Equipment : Wiring Checks

1520-1 QCR has been deemed not applicable (N/A) because RTd's wires have been soldered (As such continuity and meggering checks cannot be conducted)

NCR #: N/A

Measurement Instruments	T°	Measured by:	14-Mar-13
N/A	Before:		
	After:	ANDRITZ Rep. Kirittopoulos K.	14-Mar-2013
		Cust. Repres.:	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

RTD 3 wires

Q-1500-100

Revision # 01

Page 1 of 1

ITP step #: 1520-2

Unit : 4

Project: GM Shrum - BC

Readings in : ohms

System: Rotor pole

Brand/model:

Type: Platium 100

Formula: Tc=(R-100)/C

Coefficient (C): 0.389

Res 100

Tolerance(°C): 0.12@0°C

Instrument Tag:	Tm [°C]	RESISTANCE			Tc [°C]	VARIATION [°C]
		Rm1(A/B)	Rm2(B/C)	R		
TE001	19.9	108.4	0.6	107.8	20.1	0.2
TE002	19.7	108.4	0.7	107.7	19.8	0.1
TE003	19.9	108.4	0.6	107.8	20.1	0.2
TE004	19.4	108.3	0.7	107.6	19.5	0.1
TE005	19.3	108.2	0.7	107.5	19.3	0.0
TE006	20.0	108.3	0.6	107.7	19.8	0.2
TE007	20.1	108.4	0.6	107.8	20.1	0.0
TE008	20.2	108.4	0.6	107.8	20.1	0.1

☑ The instrument installation is sound and correct to drawings.

Shield a correctly connected at the junction boxes

NCR #: N/A

Measurement Instruments	T°	Measured by:		dd-mmm-yyyy
multimeter #2217-20	Before: 21 °C		George Barr	
Temperature gun #N10140	After: 21 °C	ANDRITZ Rep.:	Renaud Marcoux eng.	17-Jan-2013
		Cust. Repres.:	Abdul Mahmood	17-Jan-2013

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

Air gap verification - Vibro-Meter LS-120

Q-1500-430F

ITP step #: 1520-3

Revision # 00

Project: GM Shrum

Unit : 4

Page 1 of 1

System:

Brand: Vibro-Meter

Model: LS-120

Instrument Tag:	Gap/Output variation	
	Yes	No
Pole 37 Top		4.956vdc
Pole 1 Top		4.946vdc
Pole 1 Bottom		5.993vdc

Note : No acces to the airgap sensors as shrouds already installed.

☐ The instrument installation is sound and correct to drawings.

NCR #:

Measurement Instruments	T°	Measured by:		dd-mmm-yyyy
Multimeters	Before: 21°C		George Barr	
	After: 21°C	ANDRITZ Rep.:	Renaud Marcoux eng.	17-Jan-2013
		Cust. Repres.	Abdul Mahmood	17-Jan-2013

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

Revision # 00

ITP step #: 1520-5

Unit : 4

Project: GM Shrum

Equipment : Final G4 Rotor Inspection prior to commissioning

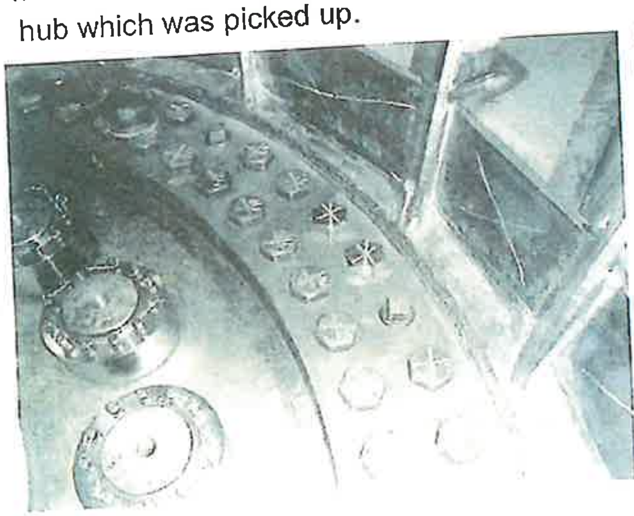
A final rotor visual inspection was carried on January 17th with 2 BC Hydro employees alongside with Andritz representative.

The inspection was focused inside the rotor as there was no access to the top of the rotor as the cover plates had previously been installed by others.

The general cleanliness of the rotor was satisfatory. The areas inspected consisted of the rotor hub, the 20 rotor spider pockets, the brake track ledge, and the rotor rim air ducts.

No anomalies were observed. No left-over tooling or material was discovered.

Few foreign debris were observed. There were few shavings observed in the air ducts which were removed with the use of a magnet. There was a piece of tape on the floor on the rotor hub which was picked up.



NCR #:

Measurement Instruments	T°		dd-mmm-yyyy
Visual	Before: N/A	Measured by: L.Luchisnki T.Quibell	17-Jan-2012
	After: N/A	J-L Cote	18-Jan-2012
		ANDRITZ Rep.: Jean-Luc Cote	18-Jan-2012
		Cust. Repres. Abdul Mahmood	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 1520-9

Revision # 00

Project: BCH GMS

Unit : 4

Page 1 of 1

QCR Q-1522-010 ITP step 1520-10 covers the content of ITP step 1520-9 and as such all relevant information for both steps can be found in it.

NCR #: N/A

Measurement Instruments	T°	Measured by:	14-Mar-13
N/A	Before:		
	After:	ANDRITZ Rep. Kirittopoulos K.	14. Mar. 13
		Cust. Repres.:	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

Q-1522-010

Revision # 00

Mechanical Run

ITP step #: 1520-10

Unit : 4

Project: BC Hydro GMS

Unit: mils pk-pk (ex: um p-p, mils 1X)

Vibration & Balance - Shaft Relative

Initial Readings

RPM					Thrust Bearing X	Thrust Bearing Y	Turbine Bearing X	Turbine Bearing Y	1x or Total
					8.3/70	7.4/320	0.8	0.6	1x
150	Initial Runouts (Jan. 23rd, 2013)				11.1	10.6	2.3	2.5	Total
					3.6/28	3.2/284	0.92	0.5	1x
150	After balance weight (Jan. 24th, 2013)				5.2	4.5	2.9	2.9	Total
					3.08/10	2.67/267	0.41	0.62	1x
150	After 160% Overspeed (Jan. 24th, 2013)				6.22	4.5	2.9	2.7	Total

Balancing Weight Added

Location		Final Weight	
Pole #	Angle	LBs	
16		40	Spider
22		30	Rim

Setpoint Limits

	Alarm	Trip
0		
0		
Thrust Bearing	BCH	BCH
Turbine Bearing	BCH	BCH

Final Readings

RPM	0 X	0 Y	0 X	0 Y	Thrust Bearing X	Thrust Bearing Y	Turbine Bearing X	Turbine Bearing Y	1x or Total
					0.25/na	0.15/na	0.25/na	0.3/na	1x
150	Speed no load zero voltage (Feb. 19th, 2013)				1.7	2.0	2.4	2.5	Total
					.4/104	0.8/330	1.0/300	1.0/246	1x
150	Speed no load rated voltage (Feb. 19th, 2013)				2.0	2.5	4.5	5.2	Total
					0.4/106	1.2/10	1.5/325	1.9/243	1x
150	Maximum Load - 255 Mw (Feb. 19th, 2013)				1.4	2.4	2.4	2.8	Total

NCR #: ____________

Measurement Instruments	T°		
Bently Nevada ADRE	Before: na	Measured by: Winston Crawford	20/02/2013
Dactron Focus	After: na	ANDRITZ Rep.: Jean-Luc Cote	
Andritz Hydro Equipment		Cust. Repres.: Yutao Zhang	[signature] 20/02/2013

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

Q-1522-010

Revision # 00

Mechanical Run

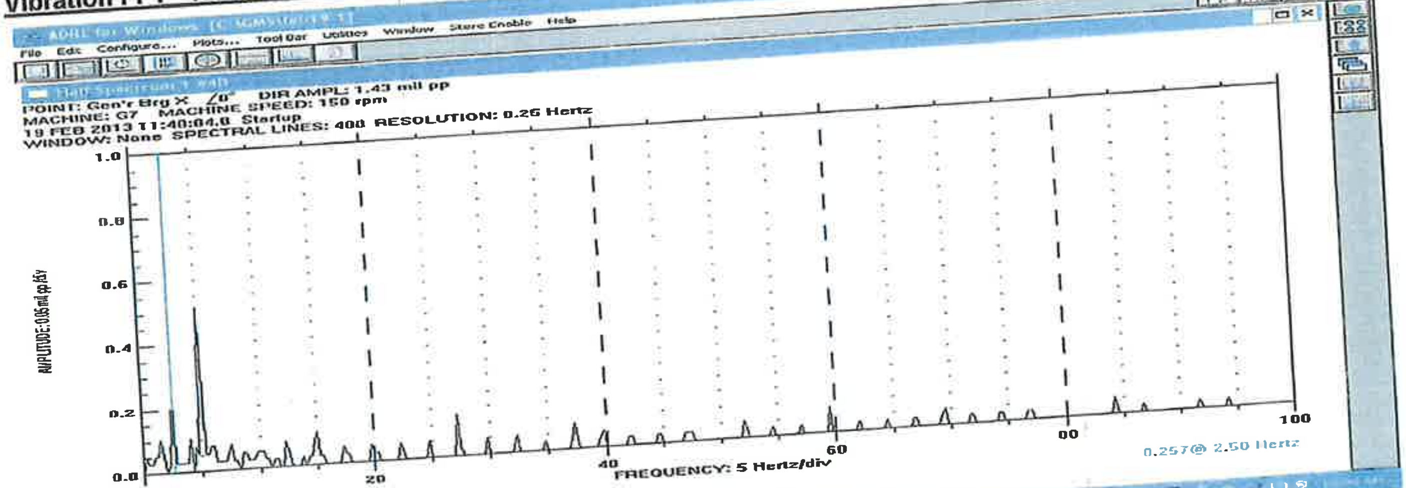
ITP step #: 1520-10

Unit : 4

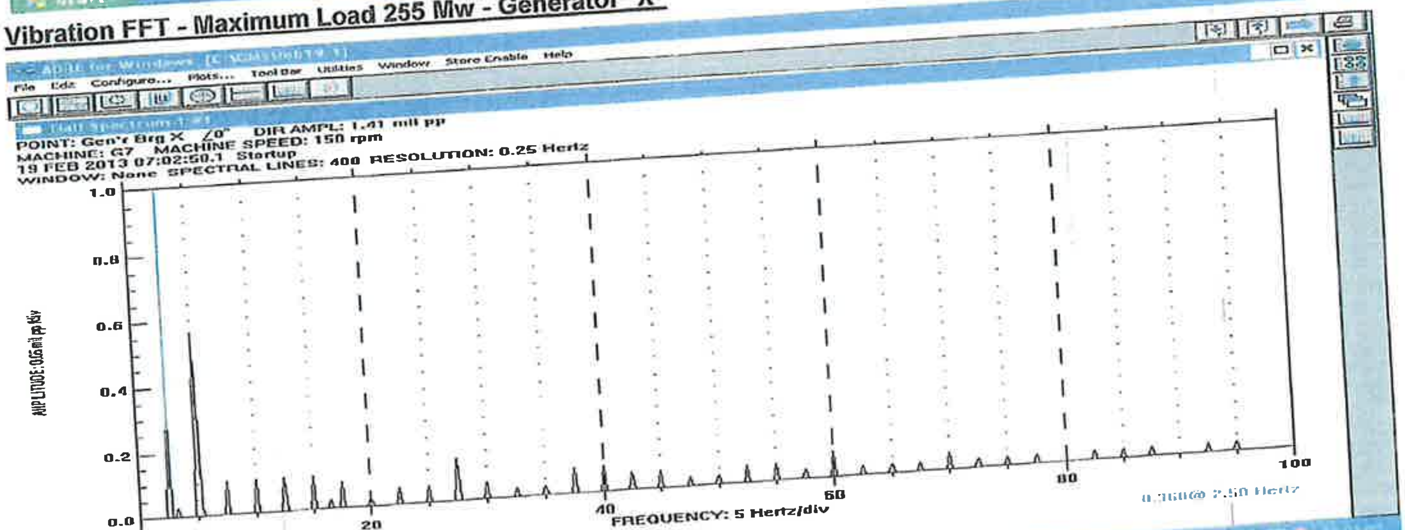
Project: BC Hydro GMS

Vibration & Balance - Shaft Relative

Vibration FFT - Nominal speed - Generator "X"



Vibration FFT - Maximum Load 255 Mw - Generator "X"



Measurement Instruments	T°		
Bently Nevada ADRE	Before: na	Measured by: Winston Crawford	20/02/2013
Dactron Focus	After: na	ANDRITZ Rep.: Jean-Luc Cote	
Andritz Hydro Equipment		Cust. Repres.: Yutao Zhang	20/02/2013

Mechanical Run

Q-1522-010

Revision # 00

Page 3 of 8

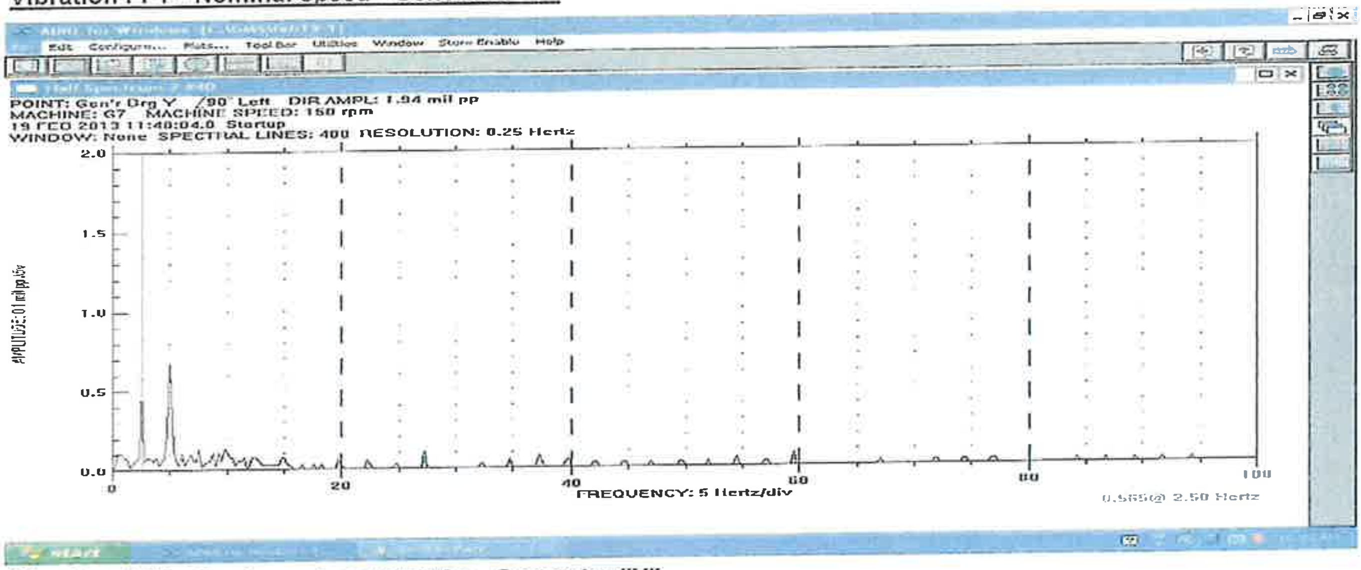
ITP step #: 1520-10

Project: BC Hydro GMS

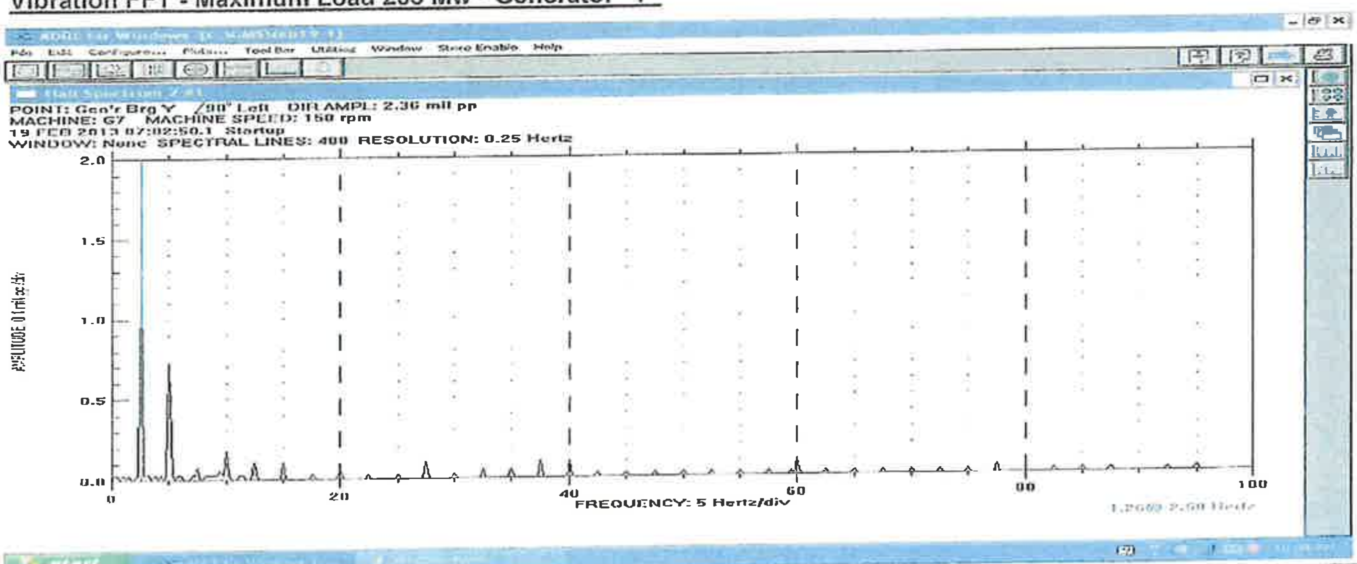
Unit : 4

Vibration & Balance - Shaft Relative

Vibration FFT - Nominal speed - Generator "Y"



Vibration FFT - Maximum Load 255 Mw - Generator "Y"



Measurement Instruments	T°		Measured by:	Winston Crawford	20/02/2013
Bently Nevada ADRE	Before:	na			
Dactron Focus	After:	na	ANDRITZ Rep.:	Jean-Luc Cote	
Andritz Hydro Equipment			Cust. Repres.:	Yutao Zhang	20/02/2013

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

Q-1522-01

Revision # 00

Mechanical Run

ITP step #: 1520-10

Unit : 4

Project: BC Hydro GMS

Vibration & Balance - Bracket absolute

Unit: in/s 0-pk (ex: mm/s 0-p, in/s 0-pk)

Initial Readings

RPM					Thrust Bearing		Turbine Bearing		1x or Total
					X	Y	X	Y	
					0.002	0.002	0.006	0.02	1x
150	Speed no load zero voltage								Total
					0.002	0.003	0.002	0.003	1x
150	Maimum load - 255 Mw								Total
									1x
									Total

Setpoint Limits

	Alarm in/s 0-pk	Trip in/s 0-pk
0		
0		
Thrust Bearing	na	na
Turbine Bearing	na	na

Final Readings

RPM	0		0		Thrust Bearing		Turbine Bearing		1x or Total
	X	Y	X	Y	X	Y	X	Y	
					0.002	0.002	0.006	0.02	1x
150	Speed no load zero voltage								Total
					0.002	0.003	0.002	0.003	1x
150	Maimum load - 255 Mw								Total
									1x
									Total

NCR #: ____________

Measurement Instruments	T°	
Bently Nevada ADRE	Before:	na
Dactron Focus	After:	na
Andritz Hydro Equipment		

Measured by: Winston Crawford 20/02/2013

ANDRITZ Rep.: Jean-Luc Cote

Cust. Repres. Yutao Zhang 21/02/2013

Installation and Commissioning

Quality Control Record

Q-1522-010

Revision # 00

Mechanical Run

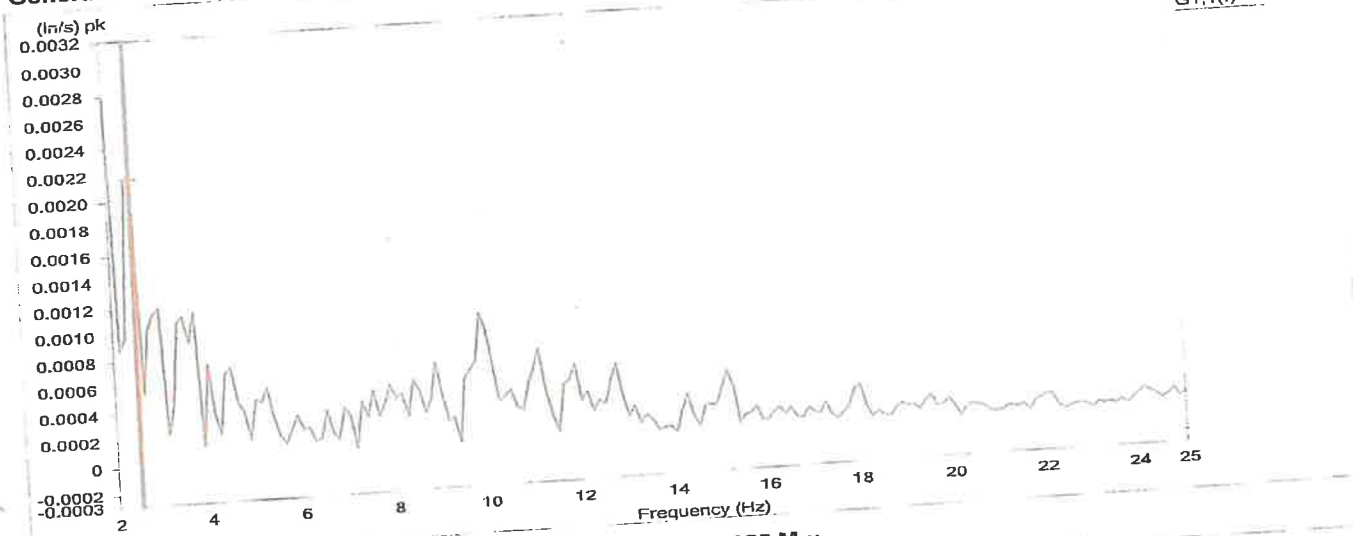
ITP step #: 1520-10

Unit : 4

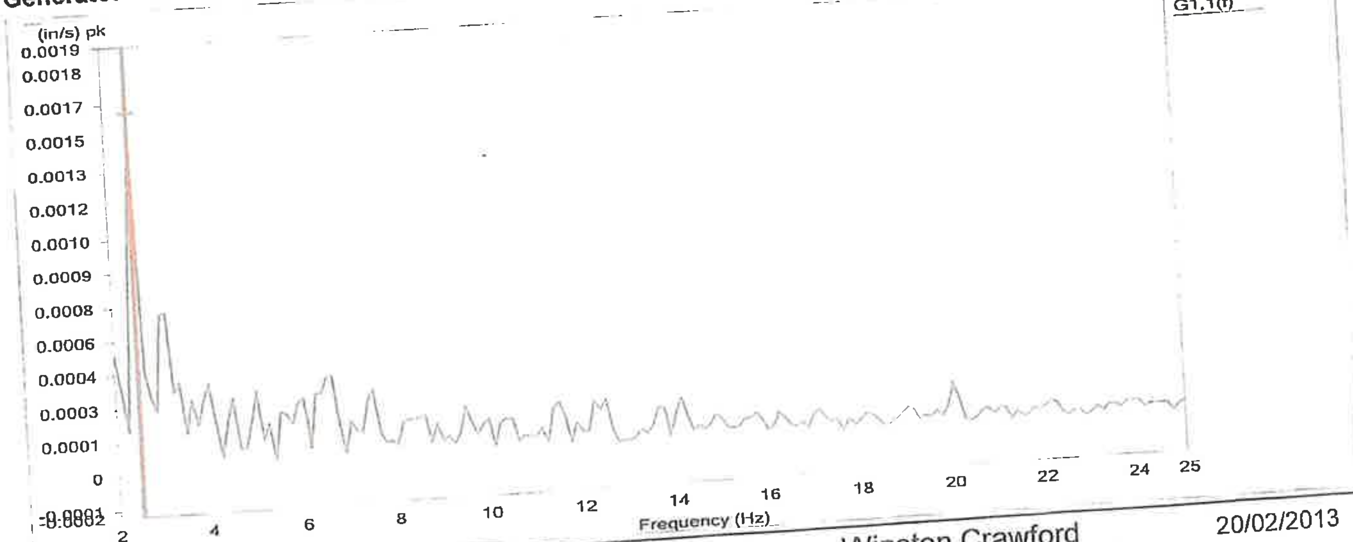
Project: BC Hydro GMS

Vibration & Balance - Bracket absolute

Generator Bearing Bracket Vibration "X" - speed no load zero voltage



Generator Bearing Bracket Vibration "X" - Maximum Load 255 Mw



Measurement Instruments	T°	Measured by: Winston Crawford	20/02/2013
Bently Nevada ADRE	Before: na		
Dactron Focus	After: na	ANDRITZ Rep.: Jean-Luc Cote	
Andritz Hydro Equipment		Cust. Repres. Yutao Zhang	26/02/2013

Installation and Commissioning

Quality Control Record

Mechanical Run

Q-1522-010

Revision # 00

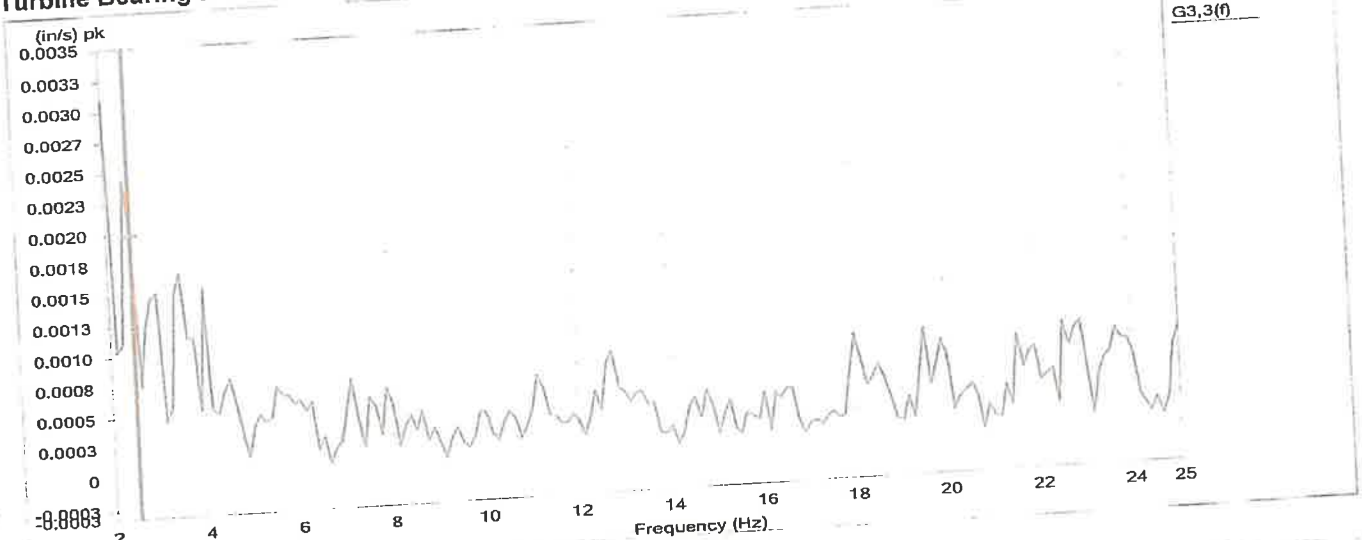
ITP step #: 1520-10

Unit : 4

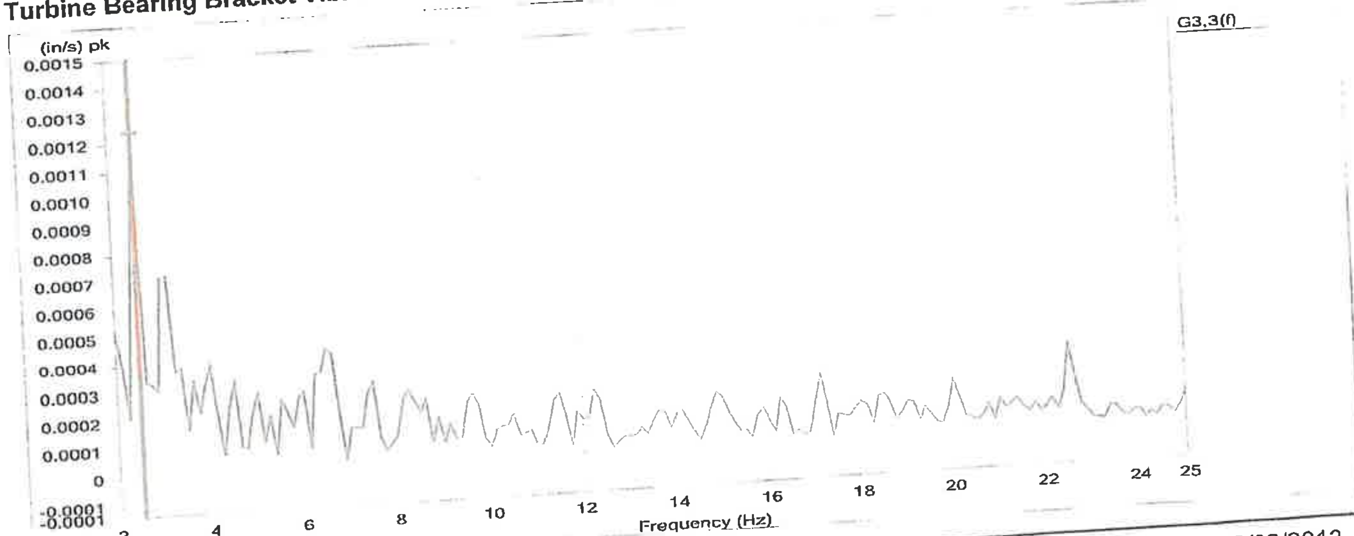
Project: BC Hydro GMS

Vibration & Balance - Bracket absolute

Turbine Bearing Bracket Vibration "X" - speed no load zero voltage



Turbine Bearing Bracket Vibration "X" - Maximum Load 255 Mw



Measurement Instruments	T°		
Bently Nevada ADRE	Before: na	Measured by: Winston Crawford	20/02/2013
Dactron Focus	After: na	ANDRITZ Rep.: Jean-Luc Cote	
Andritz Hydro Equipment		Cust. Repres. Yutao Zhang	26/02/2013

Installation and Commissioning

Quality Control Record

Mechanical Run

ITP step #: 1520-10

Project: BC Hydro GMS

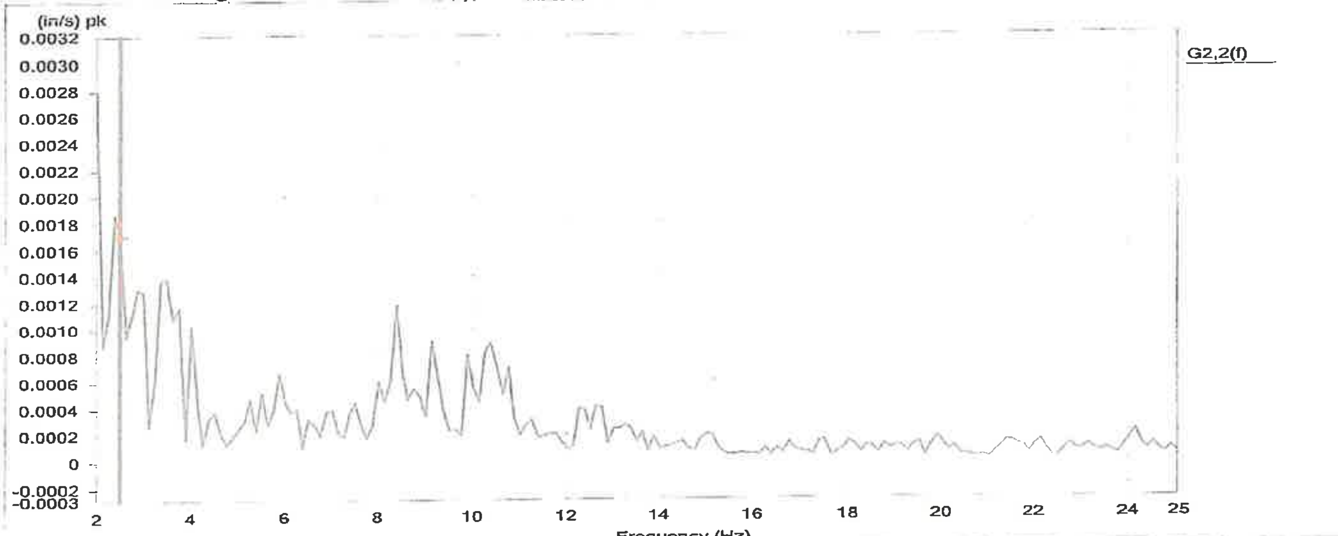
Unit : 4

Q-1522-010

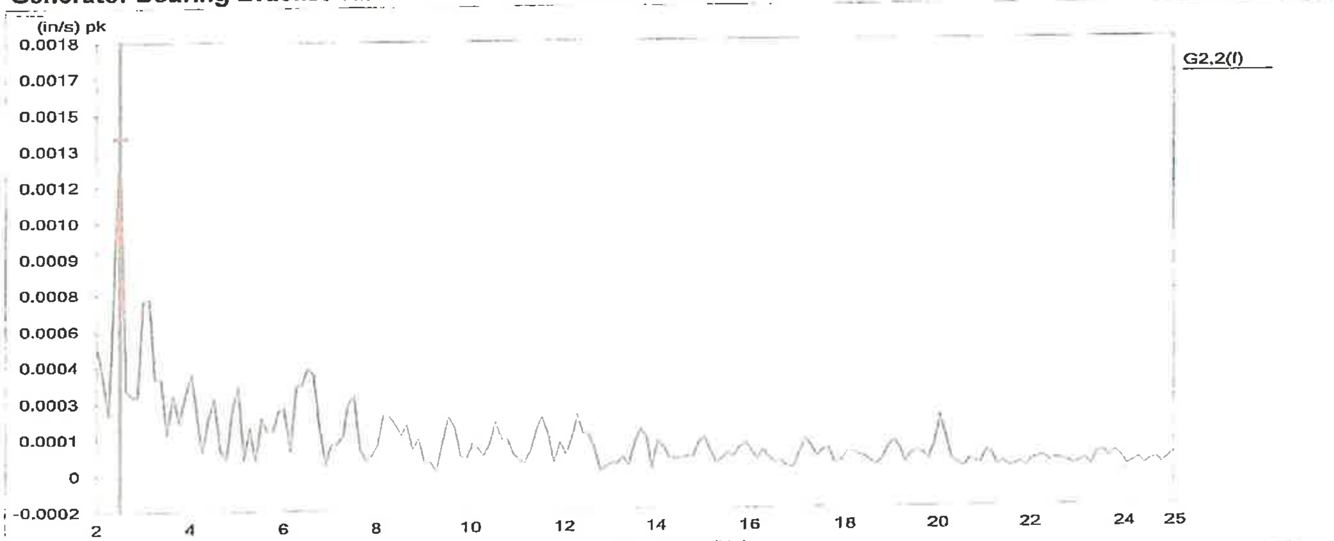
Revision # 00

Vibration & Balance - Bracket absolute

Generator Bearing Bracket Vibration "Y" - speed no load zero voltage



Generator Bearing Bracket Vibration "Y" - Maximum Load 255 Mw



Measurement Instruments		T°	Measured by:	Winston Crawford	20/02/2013
Bently Nevada ADRE	Before:	na			
Dactron Focus	After:	na	ANDRITZ Rep.:	Jean-Luc Cote	
Andritz Hydro Equipment			Cust. Repres.	Yutao Zhang	26/02/2013

Mechanical Run

Q-1522-010

ITP step #: 1520-10

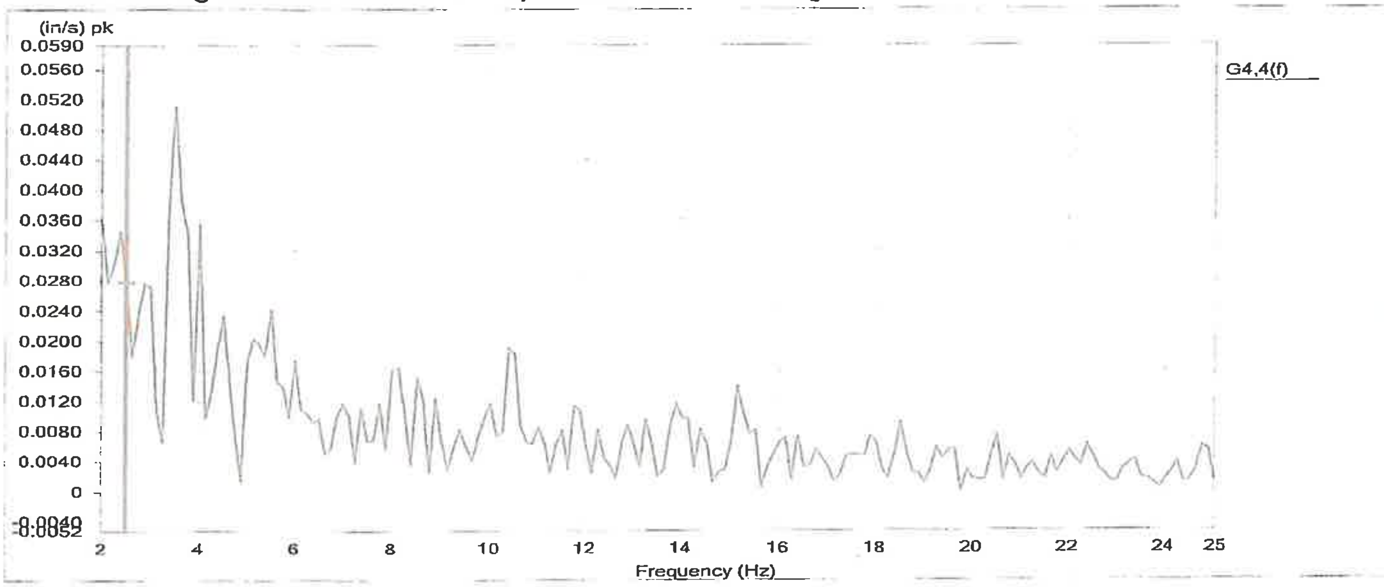
Revision # 00

Project: BC Hydro GMS

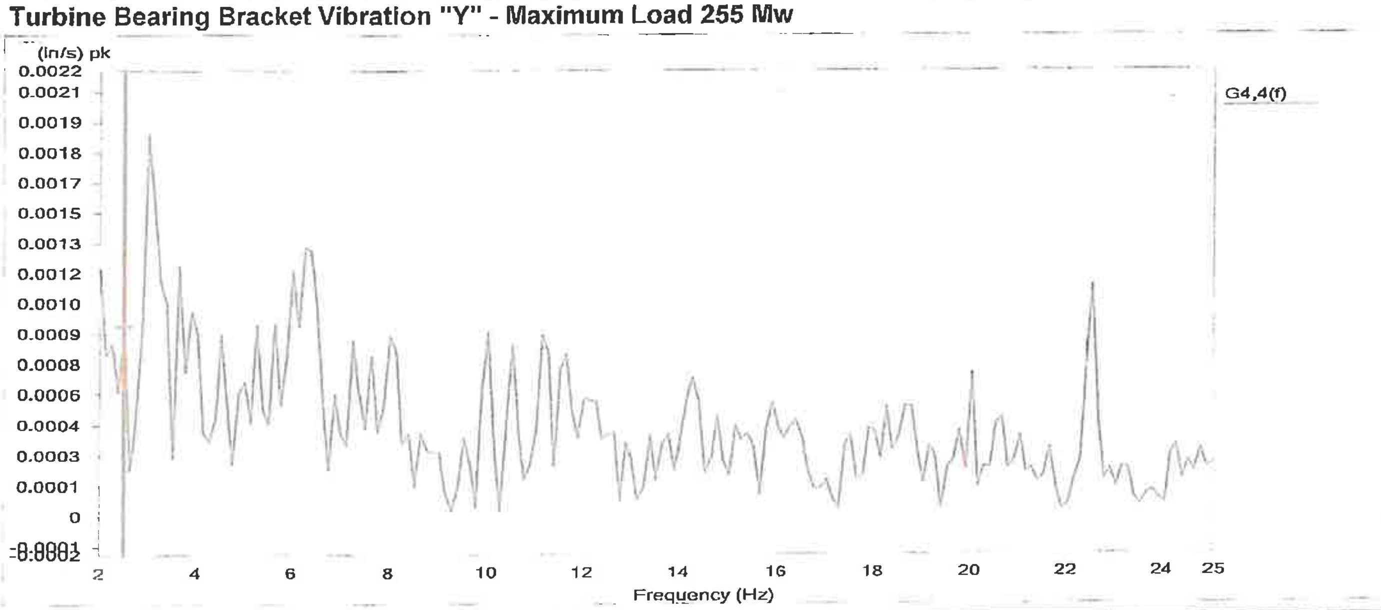
Unit : 4

Vibration & Balance - Bracket absolute

Turbine Bearing Bracket Vibration "Y" - speed no load zero voltage



Turbine Bearing Bracket Vibration "Y" - Maximum Load 255 Mw



Measurement Instruments		T°	Measured by:	Winston Crawford	20/02/2013
Bently Nevada ADRE	Before:	na			
Dactron Focus	After:	na	ANDRITZ Rep.:	Jean-Luc Cote	
Andritz Hydro Equipment			Cust. Repres.	Yutao Zhang	26/02/2013

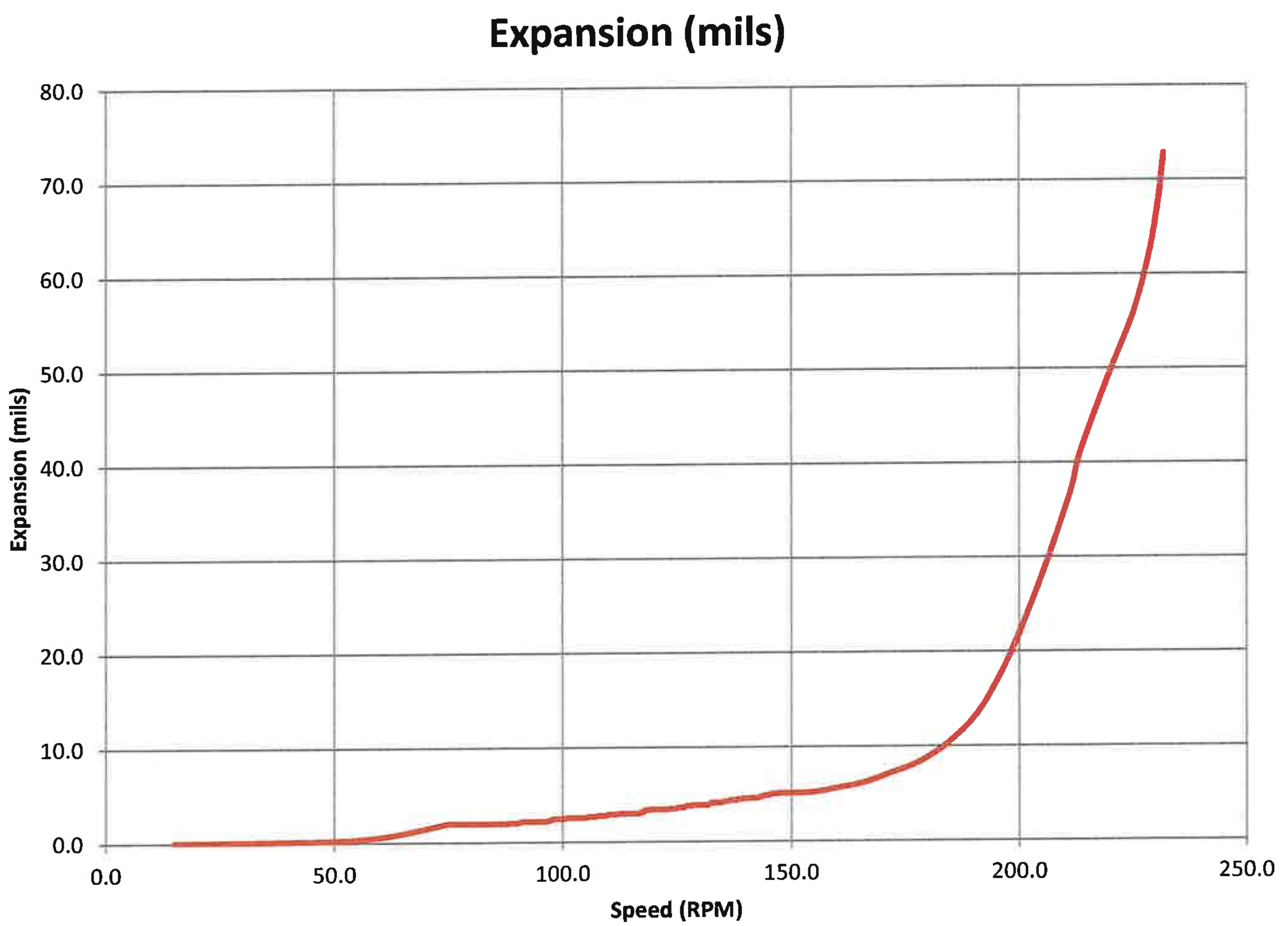
ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

Rotor Rim Expansion Measurement		**Q-1522-140**
ITP step #: 1520-11		Revision # 00
Project: BC Hydro GMS	Unit : 4	Page 1 of 2

Average Expansion in mils compare to speed (RPM)



NCR #: ____________

Measurement Instruments	T°	Measured by:	Winston Crawford	24/01/2013
Rim measurment equip.	Before: ______			
Andritz Hydro	After: ______	ANDRITZ Rep.:	Jen-Luc Cote	
(see report)		Cust. Repres.:	Yutao Zhang	

ANDRITZ Hydro

Quality Control Record

Rotor Rim Expansion Measurement Q-1522-140

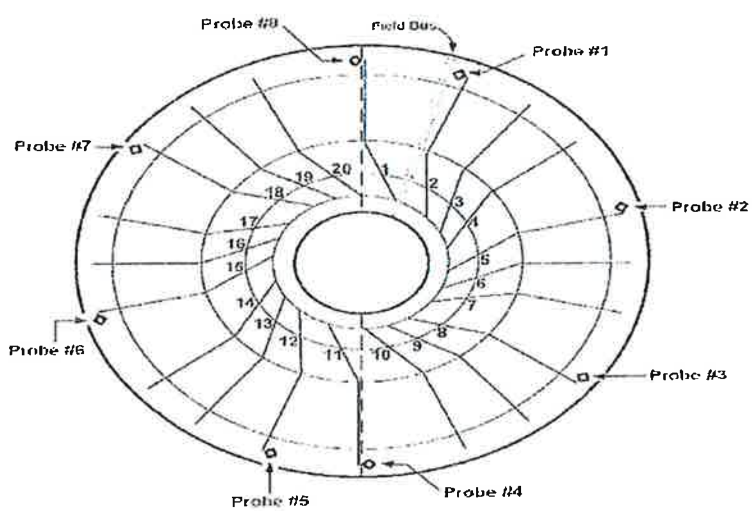
ITP Step #	1520-11			
Project	BC Hydro GMS	Unit :	4	Page 2 of 2

Procedure

Eight proximity probes were mounted on the rotor spider arms
(generally equally spaced) as shown on the sketch below.
These probes were mounted axially just above the rotor spider main disk
The proximity probes were oriented in the radial direction and capable of measuring any change in the gap between the spider and the rim.
The unit was rotated over a speed range from standstill to 231.9 rpm (approximately 155% of rated speed).
The probe tip-to-rim gap was recorded for all probes and the averaged data was plotted.

Equipment

Probes - Bently Nevada 3300 series 8 mm proximitor sensors (Andritz Equipment)
Data Loggers - ACR Owl DC voltage loggers (Andritz Equipment)
Batteries - Custom batteries to power proximitor drivers



Measured by:	Winston Crawford	24/01/2013
		Revised - May 8th, 2013
ANDRITZ Rep.	Jean-Luc Cote	By - W.G. Crawford
Cust. Repres.:	Yutao Zhang	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 1520-12

Revision # 00

Project: BC Hydro GMS

Unit : 4

Page 1 of 1

Equipment : Visual Inspection of Rotating Parts

BCH and Andritz Hydro - inspection of rotor keys and spider after 165% overspeed test.

The rotor pole driven keys were re-pounded to refusal after oversped. No anomalies were observed during inspection completed after 165% overspeed.

NCR #:

Measurement Instruments	T°	Measured by: BCH / Andritz Hydro Team	1/28/2013
Visual	Before:		
	After:	ANDRITZ Rep.: Jean-Luc Cote	
		Cust. Repres. Yutao Zhang	

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

GMS - Generator Static test

Q-1510-030

ITP step #: 1520-13

Revision # 00

Project: BCH GMS

Unit : 4

Page 1 of 1

Field Cold Resistance

*Resistance measured at collector rings

Measured Resistance [RT] @ C	0.12194
Resistance [Rs] @ 25°C	0.12346
Resistance [Rs] @ 75°C	0.14725
Temperature [T t] °C	21.8

Note - Average of 44 stator temperatures used along with rotor pole RTD's

Megger and Polarization Index (Rotor)

Time (min)	Insulation Resistance	
1	2.6	Gigohms
2	3.3	"
5	4.0	"
10	3.5	"
P index	1.4	"

Initial megger done on Feb. 1st, 2013

Repeat megger for 1 minute on Feb. 3rd, 2013

Test voltage Applied	2500 Vdc

P index = (gigohms @ 10 minute / gigohms @ 1minute)

Hipot (Rotor)

(BCH Procedures followed)

1 Minute 5 Kv ac Hipot to Ground - **Pass**

Test voltage Applied	5000 Vac

Temp - 19.2 Deg. C

Humidity - 28.4 %

NCR #: ____________

Measurement Instruments	T°	Measured by:	Winston Crawford	03/01/2013
BCH Equipment	Before: 19.2		BCH Team	Revised
Megger - S1-1054/2	After: 19.2	ANDRITZ Rep.:	Jean-Luc Cote	May 8th, 2013
Hipot - Hipotronics M9604124		Cust. Repres.:	Yutao Zhang	